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(54) **SYSTEM AND METHOD FOR PROVIDING
AN ADAPTIVE USER INTERFACE ON AN
ELECTRONIC APPLIANCE**

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G05B 19/042 (2006.01)

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(57) **ABSTRACT**

A controllable device, such as a set top box, responds to a
transmission received from a one of a plurality of controlling
devices of differing capabilities by entering into a one of a
plurality of operating modes wherein the one of the plurality
of operating modes entered into corresponds to the capa-
bilities of the controlling device from which the transmis-
sion originated.

(52) **U.S. Cl.**

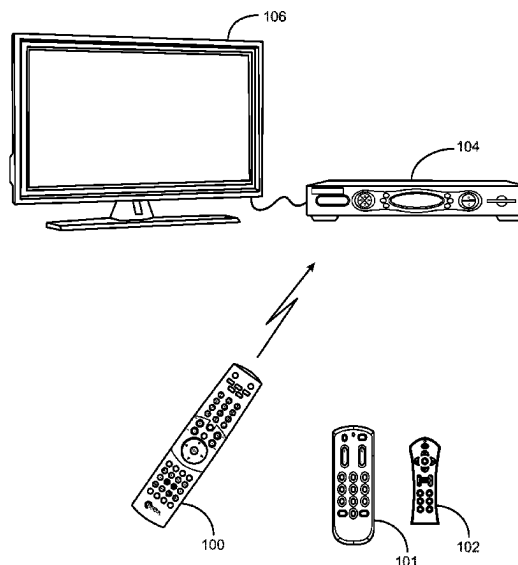
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(2013.01); **H04N 21/41265** (2020.08);

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(58) **Field of Classification Search**

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13 Claims, 9 Drawing Sheets



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- (51) **Int. Cl.**
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- (52) **U.S. Cl.**
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- (58) **Field of Classification Search**
 CPC H04N 21/42207; H04N 21/4221; H04N 21/4223; H04N 21/4305
 See application file for complete search history.

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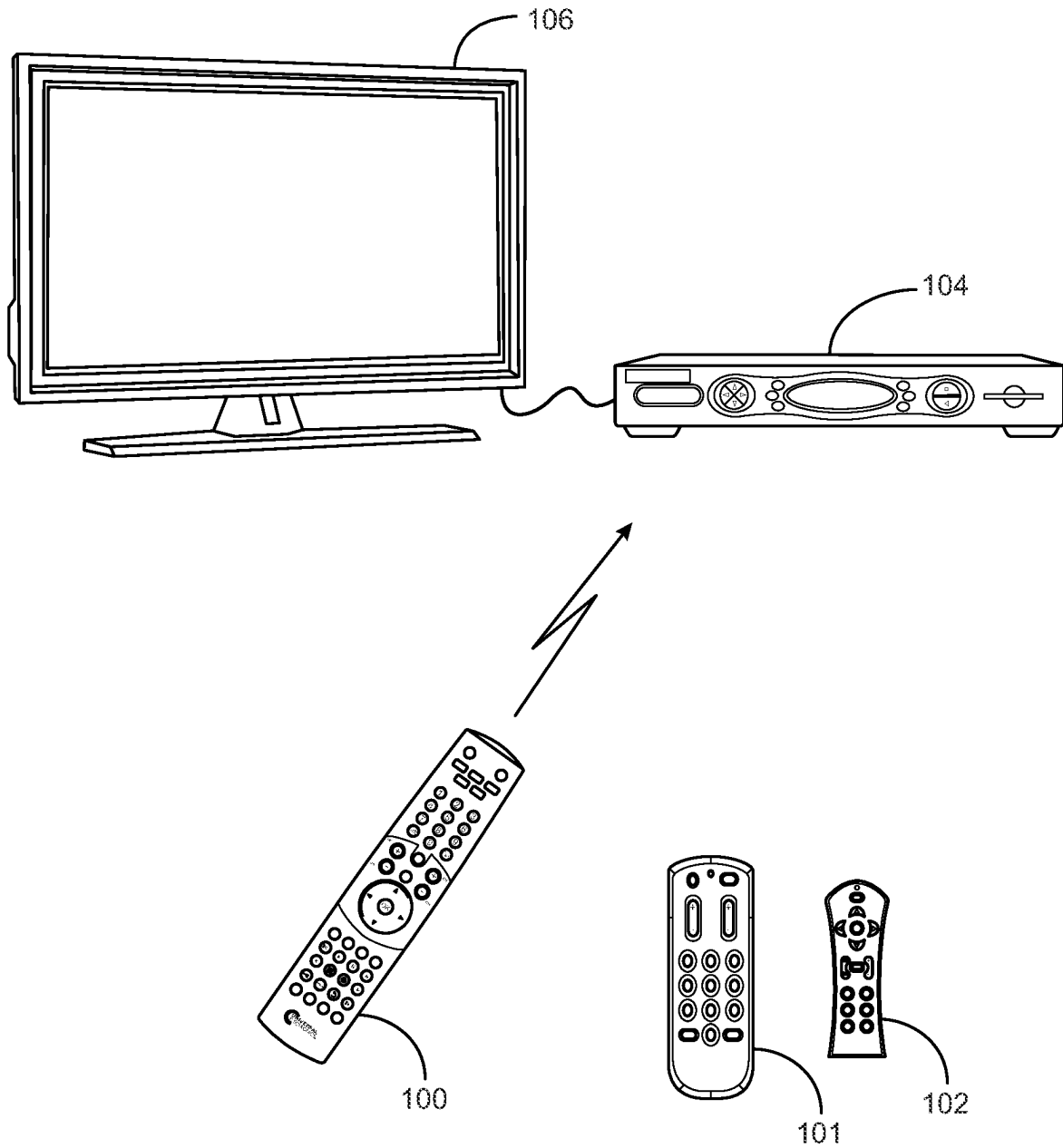


FIG. 1

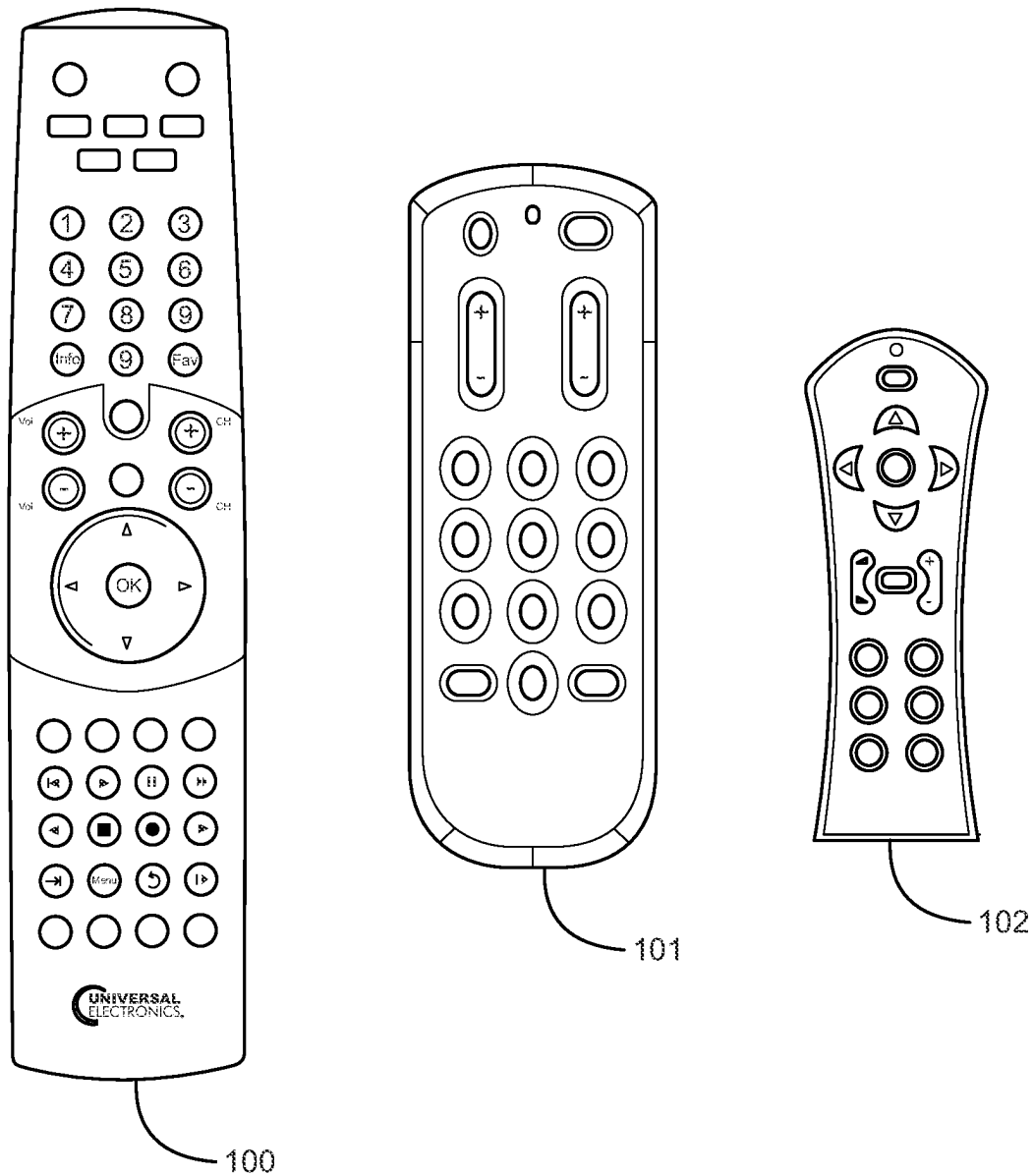


FIG. 2

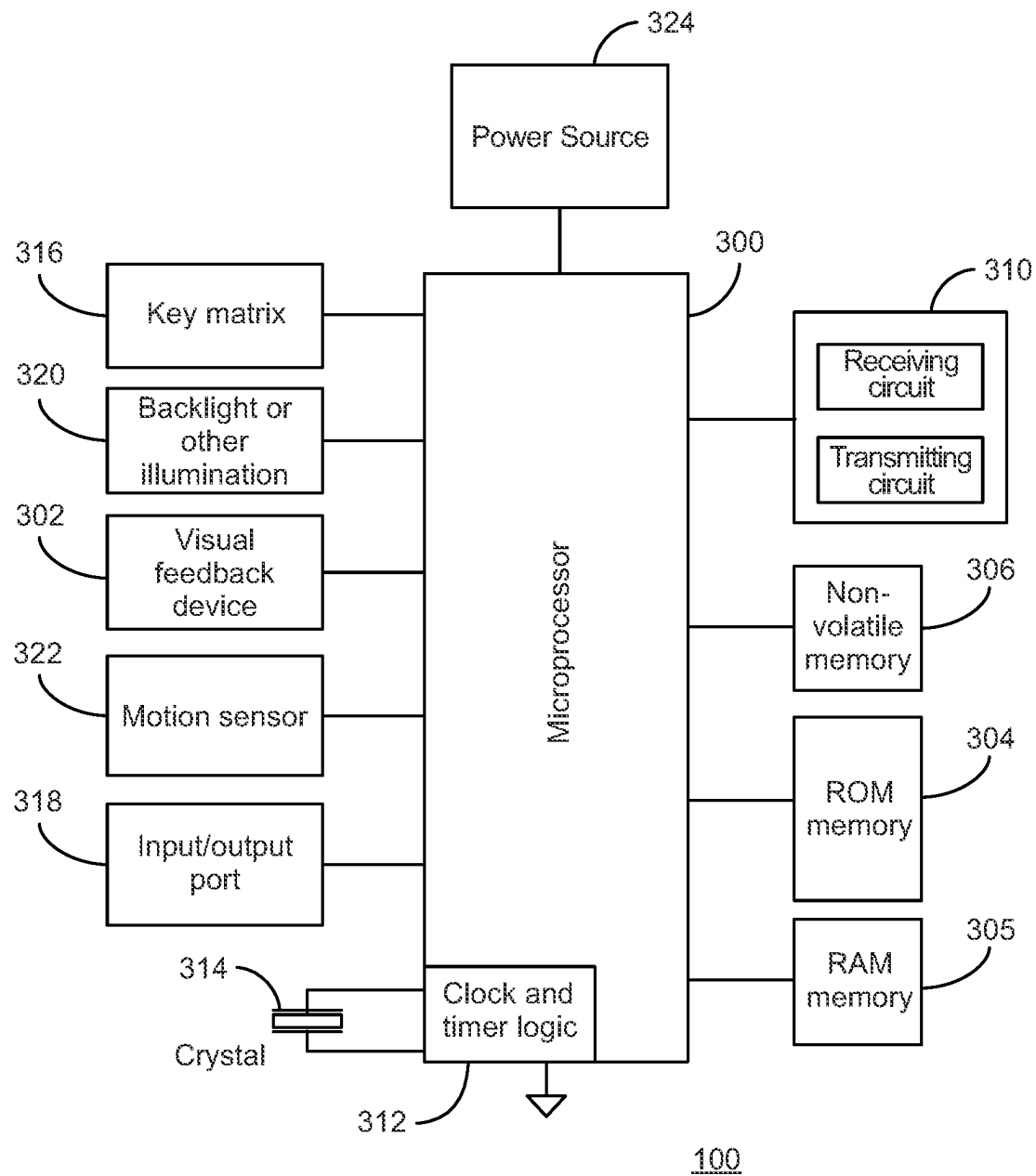


FIG. 3

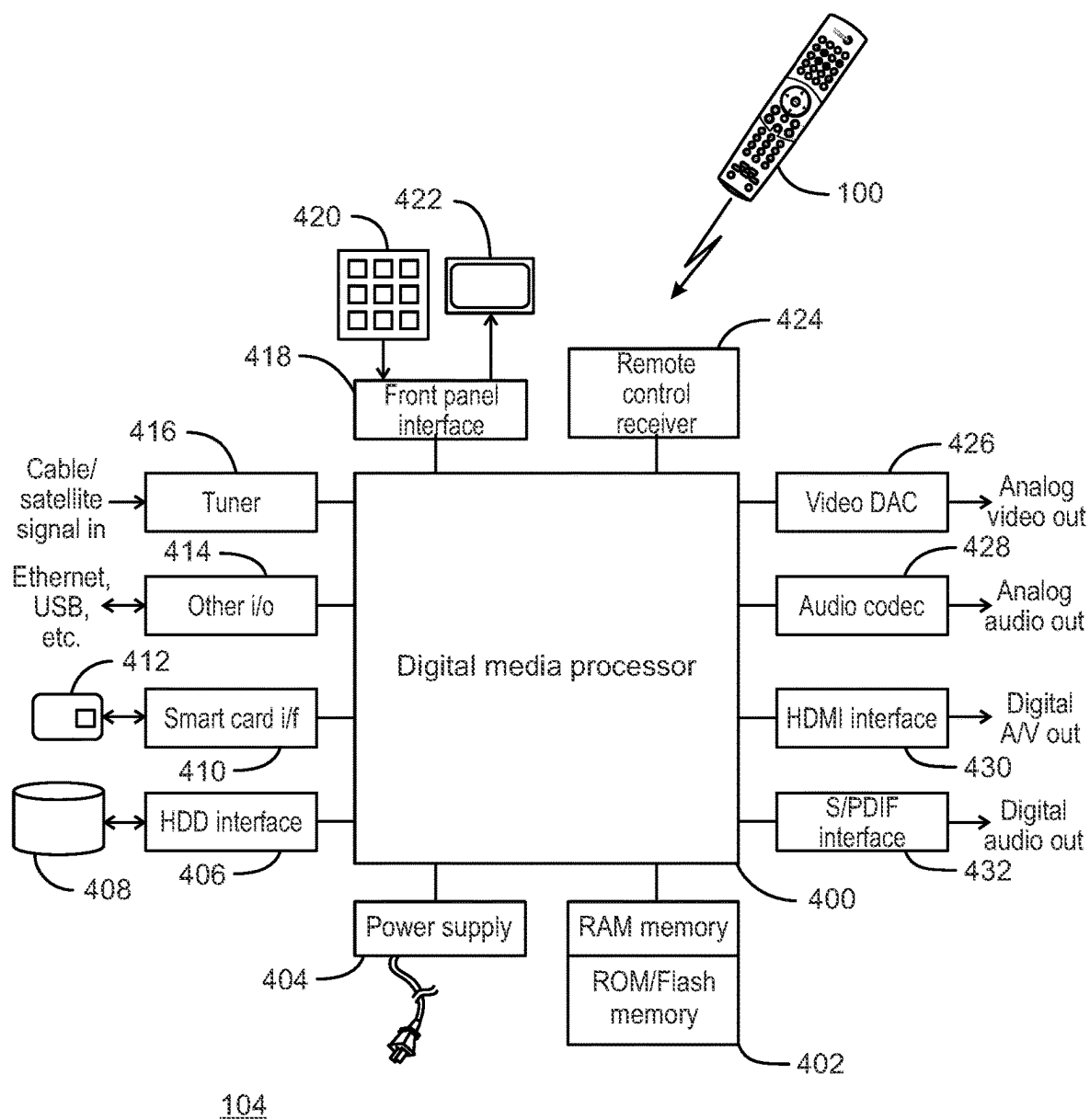


FIG. 4

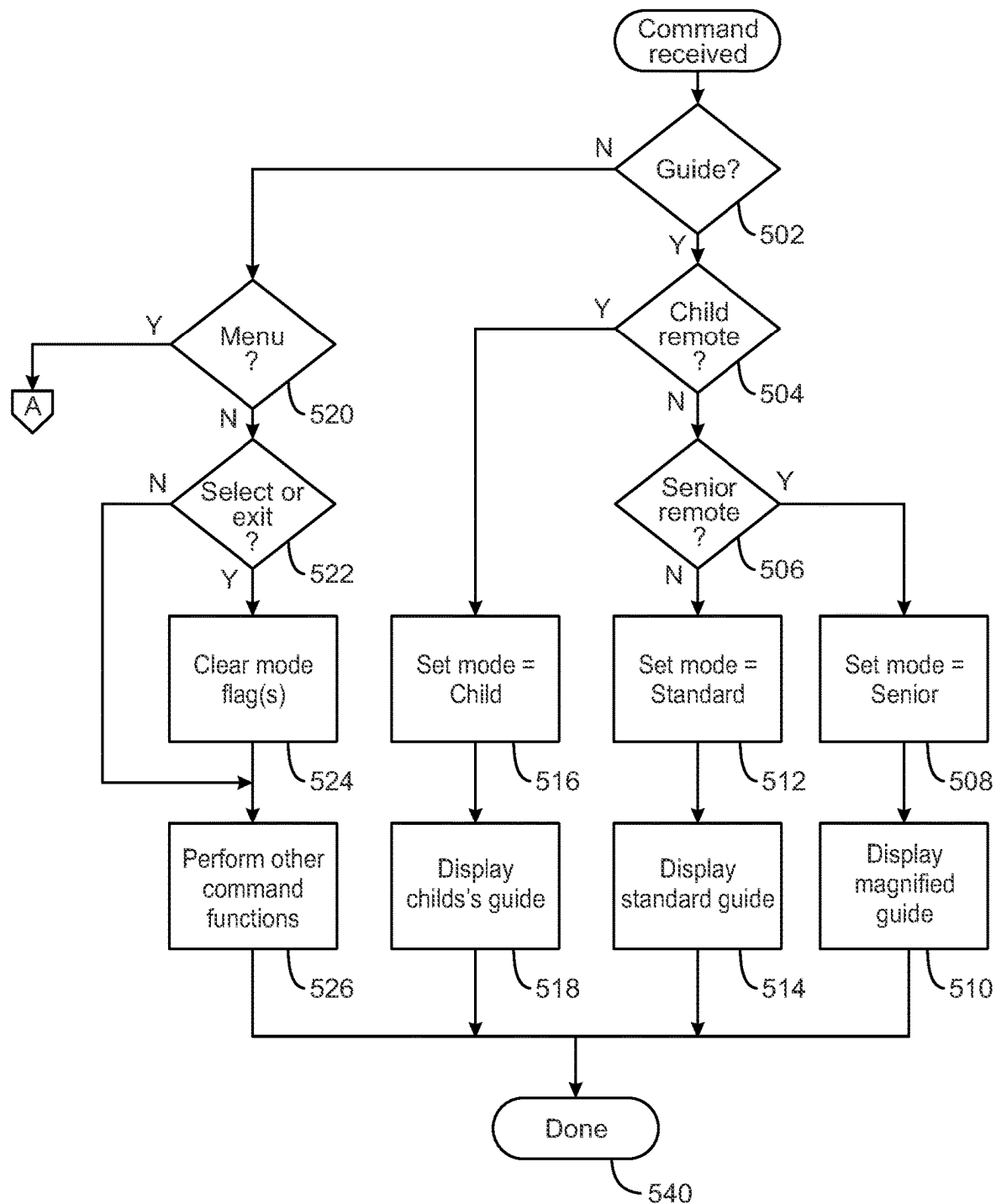


FIG. 5A

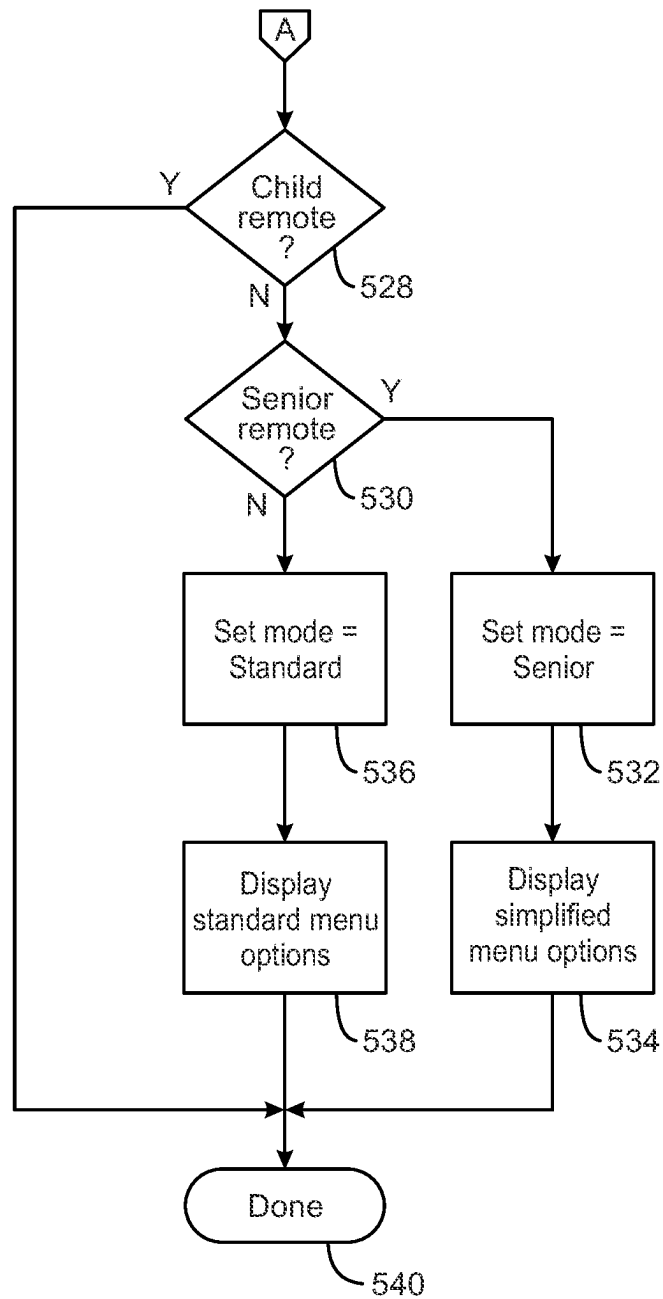


FIG. 5B

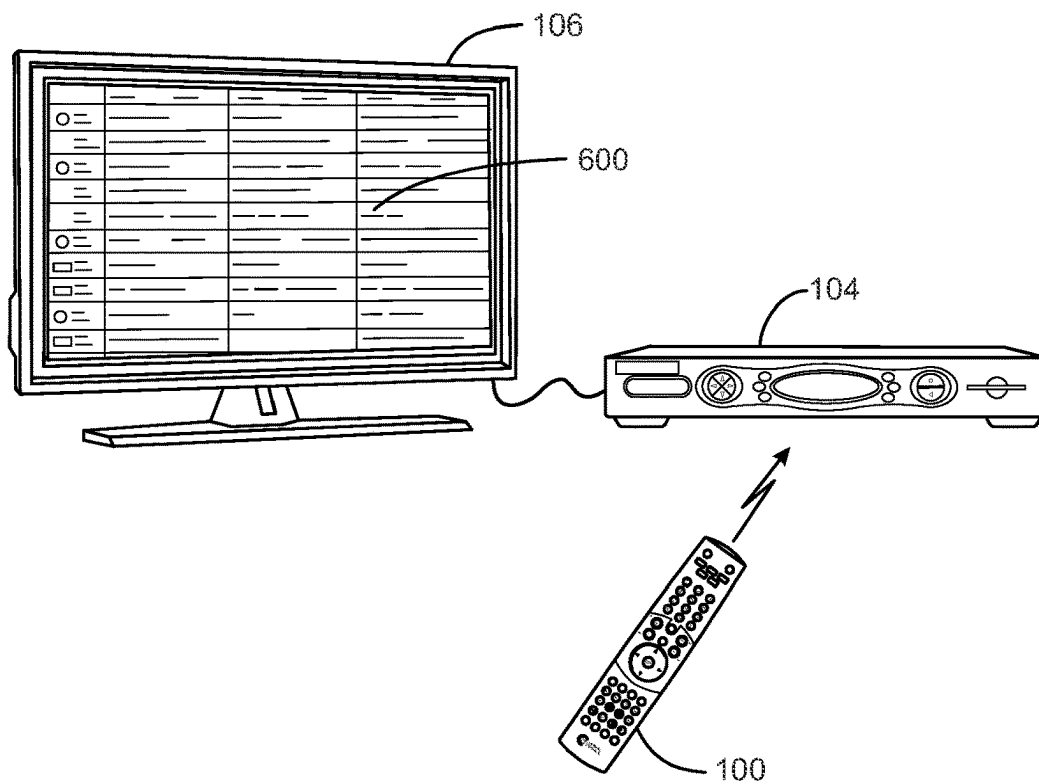


FIG. 6

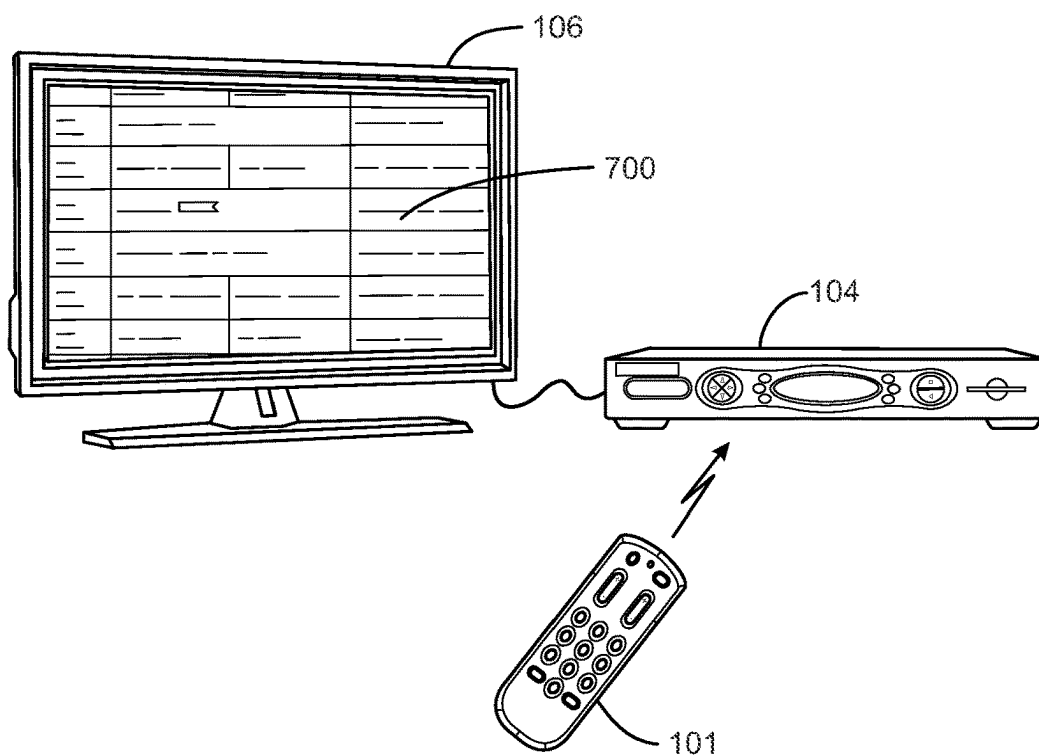


FIG. 7

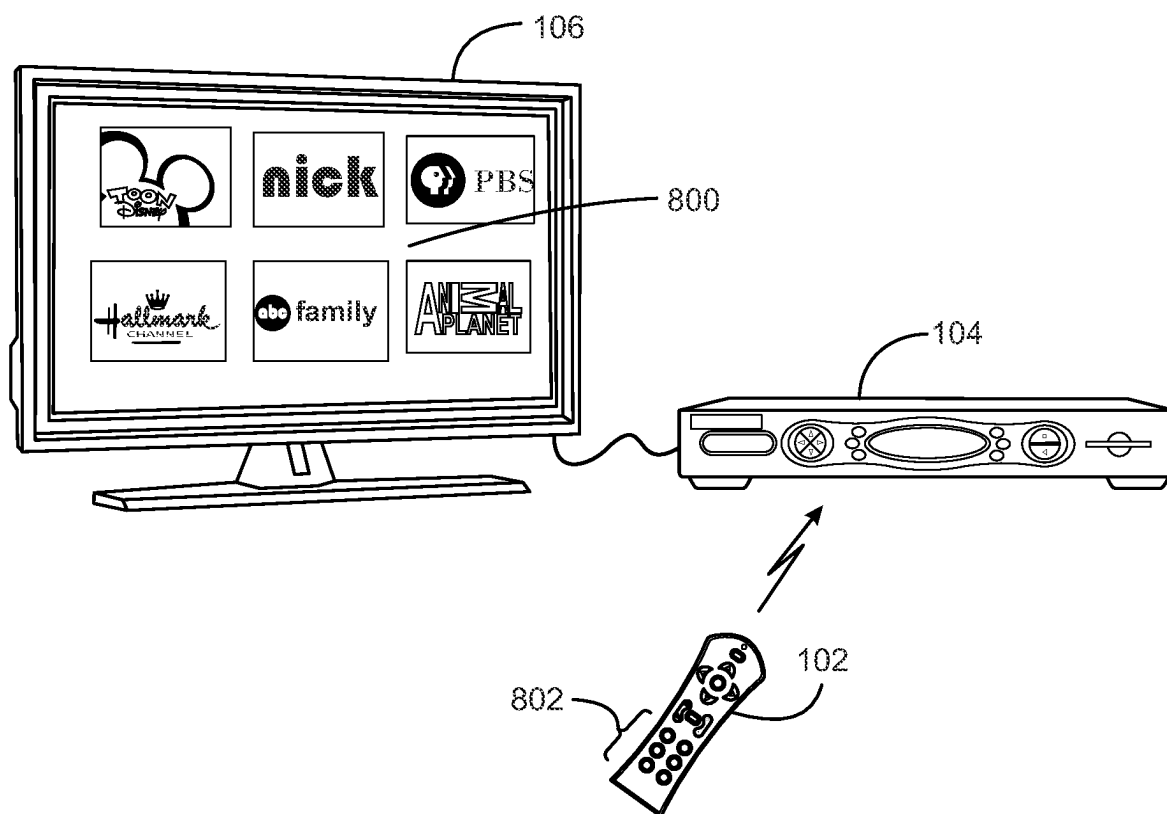


FIG. 8

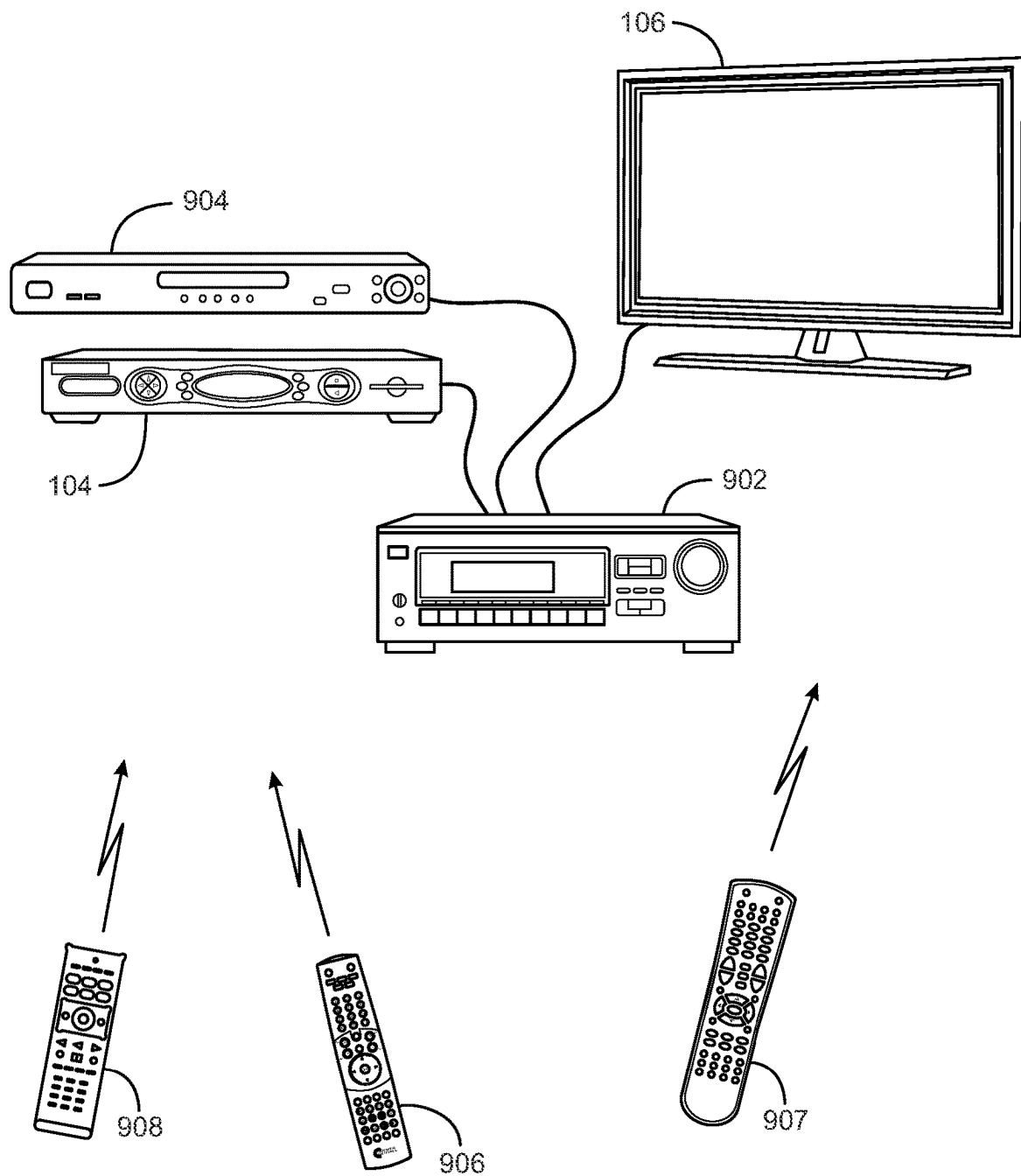


FIG. 9

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SYSTEM AND METHOD FOR PROVIDING AN ADAPTIVE USER INTERFACE ON AN ELECTRONIC APPLIANCE

RELATED APPLICATION INFORMATION

This application claims the benefit of and is a continuation of U.S. application Ser. No. 16/815,122, filed on Mar. 11, 2020, which application claims the benefit of and is a continuation of U.S. application Ser. No. 15/962,451, filed on Apr. 25, 2018, which application claims the benefit of and is a continuation of U.S. application Ser. No. 13/225,635, filed on Sep. 6, 2011, which application claims the benefit of U.S. Provisional Application No. 61/380,860, filed Sep. 8, 2010. The disclosure within these applications is incorporated herein by reference in their entirety.

BACKGROUND

Consumer electronic appliances such as set top boxes for receiving and decoding cable and satellite signals, televisions, DVD players, game systems, etc. are well known in the art, as are the hand held controlling devices, for example remote controls, which may be provided for use in issuing commands to these appliances. In some instances these controlling devices may be made available in various forms, for example coded to assist in recognition of individual users in a household; optimized for a specific function such as game playing, web browsing, or movie watching; or specially adapted to the needs of the elderly or of children. While such controlling devices may serve the purposes for which they were intended, in some instances it would be advantageous if the controlled appliance were further enabled to recognize the nature of the controlling device currently in use and adapt its appliance configuration, menuing system, GUI, etc. accordingly

SUMMARY OF THE INVENTION

This invention relates generally to systems in which more than one controlling device may be available for the operation of an appliance or group of appliances, and in which different controlling devices may each be enabled to convey a specific identity to a controlled appliance to facilitate adaptation of appliance features and/or configuration in accordance with the controlling device currently in use. As appropriate for a particular embodiment, such a controlling device identity may take the form of a data field embedded within a conventional command transmission frame or may comprise a separate, discrete transmission. Additionally, according to the particular embodiment the controlling device identity may be communicated in conjunction with every command transmission; communicated only in conjunction with certain commands or keystrokes (for example menu initiation); communicated in conjunction with detected events such as a user picking up a controlling device; or a combination thereof.

In response to a received controlling device identity, a target appliance may adjust its GUI display, offer a different set of menu choices, reconfigure its mode of operation or functionality, etc. For example, as presented hereafter by way of illustrative embodiment, a set top box may be adapted to present different versions of an on-screen program guide GUI, each optimized to the population for which the controlling device currently in use is particularly intended, viz: Regular users, elderly users, or children.

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A better understanding of the objects, advantages, features, properties and relationships of the invention will be obtained from the following detailed description and accompanying drawings which set forth illustrative embodiments and which are indicative of the various ways in which the principles of the invention may be employed.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding of the various aspects of the invention, reference may be had to preferred embodiments shown in the attached drawings in which:

FIG. 1 illustrates an exemplary system in which multiple controlling devices and a set top box configured in accordance with the instant invention may be utilized;

FIG. 2 further illustrates the exemplary controlling devices of the system of FIG. 1;

FIG. 3 illustrates a block diagram of exemplary components of an exemplary controlling device of FIG. 1;

FIG. 4 illustrates a block diagram of exemplary components of the exemplary set top box of FIG. 1;

FIGS. 5a and 5b illustrate the steps of an exemplary method for adapting the operation of the set top box of FIG. 1 in accordance with the controlling device currently in use;

FIG. 6 illustrates the exemplary operation of the set top box of FIG. 1 when used in conjunction with a first exemplary controlling device;

FIG. 7 illustrates the exemplary operation of the set top box of FIG. 1 when used in conjunction with a second exemplary controlling device;

FIG. 8 illustrates the exemplary operation of the set top box of FIG. 1 when used in conjunction with a third exemplary controlling device; and

FIG. 9 illustrates a second exemplary system in which multiple controlling devices and an appliance configured in accordance with the instant invention may be utilized.

DETAILED DESCRIPTION

Turning now to FIG. 1, an exemplary system is illustrated wherein several controlling devices **100**, **101**, **102** may be provided for the purpose of commanding the operation of a digital set top box ("STB") **104**. As is known in the art, controlling devices **100** through **102** may be capable of transmitting commands to the STB using any convenient IR, RF, Point-to-Point, or networked protocol, to cause the STB to perform operational functions. As is also known in the art, one or more of controlling devices **100** through **102** may comprise so-called "universal" functionality, that is be equipped with a library of command code sets suitable for controlling multiple appliances of different manufacture, for example to enable control of the sound volume of a TV set **106** being used to render the audio/video output stream originating from STB **104**. With reference to FIG. 2, while all being capable of commanding the operation of STB **104**, each of the controlling devices may be functionally, visually, and/or ergonomically optimized for a particular user population. For example, controlling device **100** may comprise a generic or standard controlling device, shipped with every unit of STB **104** and capable of controlling all available functions of the STB including set-up, configuration, etc., and intended for use by regular and/or technically adept users. In contrast, controlling device **101** may comprise fewer, larger buttons which function to control only the everyday functionality of STB **104**, and may be intended to simplify access to STB-provided services by elderly subscribers. In some embodiments, controlling device **101** may

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also incorporate an emergency alert ("panic button") for summoning assistance, as contemplated for example in co-pending U.S. patent application Ser. No. 12/956,44. Controlling device **102** may comprise even fewer buttons packaged in a visually appealing and/or brightly colored housing which may be dimensionally adapted for smaller hands, and may be intended for use by children.

Returning now to the illustrative embodiment of FIG. 1 with particular reference to FIG. 3, for use in commanding the functional operations of one or more appliances each exemplary controlling device **100** through **102** (hereafter exemplified by controlling device **100**) may include a processor **300** coupled to a ROM memory **304**, a RAM memory **305**, a key matrix **316** (e.g., hard keys, soft keys such as a touch sensitive surface overlaid on a liquid crystal (LCD), transmission circuit(s) and/or transceiver circuit(s) **310** (e.g., IR and/or RF), clock and timer logic **312** with associated crystal or resonator **314**, a power source **324** such as battery, and as required for a particular embodiment, a non-volatile read/write memory **306** such as Flash, battery backed up SRAM, etc., means **302** to provide feedback to the user (e.g., one or more LEDs, display, speaker, and/or the like), an input/output port **318** such as a serial interface, USB port, modem, Zigbee, WiFi, or Bluetooth transceiver, etc., means **320** for backlighting a LCD display or keypad, and means **322** to detect the device being touched or picked up by a user such as a tilt switch, accelerometer, touch sensor, proximity sensor, inclinometer, etc.

As will be understood by those skilled in the art, some or all of the memories **304**, **305**, **306** may include executable instructions (collectively, the controlling device program memory) that are intended to be executed by the processor **300** to control the operation of controlling device **100**, as well as data which serves to define to the operational software the necessary control protocols and command values for use in transmitting command signals to controllable appliances (collectively, the command data). In this manner, the processor **300** may be programmed to control the various electronic components within the controlling device **100**, e.g., to monitor the key matrix **316**, to cause the transmission of signals, etc. The non-volatile read/write memory **306**, for example an EEPROM, battery-backed up RAM, FLASH, Smart Card, memory stick, or the like, may additionally be provided to store setup data and parameters as necessary. While the memory **304** is illustrated and described as a ROM memory, memory **304** can also be comprised of any type of readable media, such as ROM, FLASH, EEPROM, or the like. Preferably, read/write memories **304** and **305** are non-volatile or battery-backed such that data is not required to be reloaded after battery changes. In addition, the memories **304**, **305** and **306** may take the form of a chip, a hard disk, a magnetic disk, an optical disk, and/or the like. Still further, it will be appreciated that some or all of the illustrated memory devices may be physically combined (for example, a single FLASH memory may be logically partitioned into different portions to support the functionality of memories **304** and **306** respectively), and/or may be physically incorporated within the same IC chip as the microprocessor **300** (a so called "microcontroller") and, as such, they are shown separately in FIG. 3 only for the sake of clarity.

To cause the controlling device **100** to perform an action, the controlling device **100** may be adapted to be responsive to events, such as a sensed user interaction with the key matrix **316**, detection of a triggering event via an appropriate trigger sensor, etc. In response to an event, appropriate instructions within the program memory (hereafter the "con-

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trolling device operating program") may be executed. For example, when a function key is actuated on controlling device **100**, the controlling device **100** may retrieve from the command data stored in memory **304**, **305**, **306** a command value and control protocol corresponding to the actuated function key and, where necessary, current device mode, and transmit that command to an intended target appliance, e.g., TV **106** or STB **104**, in a format recognizable by that appliance to thereby control one or more functional operations of that appliance. In accordance with the teachings of this invention, in some embodiments the transmitted command value may further incorporate a controlling device type identifier as described in greater detail hereafter. It will be appreciated that the operating program can be used not only to cause the transmission of commands and/or data to the appliances, but also to perform other operations. While not limiting, other operations that may be performed by the controlling device **100** may include displaying information/data, favorite channel setup, macro key setup, function key relocation, etc. Examples of local operations may be found in U.S. Pat. Nos. 5,481,256, 5,959,751, and 6,014,092.

In certain embodiments, controlling device **100** may be universal, that is, adaptable to issue commands to a multiplicity of appliances of different type and/or manufacture. In such cases, for selecting a set of command data to be associated with an appliance to be controlled, data may be provided to the controlling device **100** that serves to identify an intended target appliance by its type and make (and sometimes model). Such data allows the controlling device **100** to identify the appropriate command data within a preprogrammed library of command data that is to be used to transmit recognizable commands in a format appropriate for such identified appliances. Since the operation and features of such universal controlling devices are well known in the art, for the sake of brevity these will not be discussed further herein.

Turning now to FIG. 4, an exemplary STB **104** may include, as needed for a particular application, a processor **400** coupled to a memory **402** which may comprise RAM, ROM, and/or Flash memory; a power supply **404**; a hard disk (HDD) interface **406** and associated disk drive **408**, for example to support DVR functionality; a smart card and/or PCMCIA interface **410** where required to accommodate service enablement/decryption cards **412**; other input/output interfaces **414** such as USB, Ethernet, etc.; a front panel interface **418** with associated front panel keypad **420** and display means **422**; a remote control receiver or transceiver **424** for reception of signal from a controlling device **100**; analog video and audio outputs **426, 428**; digital audio/video and audio outputs **430, 432**; and one or more tuners **416** for receiving and demodulating a cable or satellite signal. As will be appreciated, in general tuner **416** may be capable of receiving both one or more digital media streams comprising program content and a so called out-of-band data stream comprising, for example, program guide information, software updates, etc.

As is known in the art, programming instructions may be stored in memory **402** which when executed by processor **400** may enable the STB appliance **104** to receive and decode digital media and data streams from tuner **416**; to store received media content and data on HDD **408** or in memory **402**; to output received (from tuner **416**) or retrieved (from HDD **408**) media content via one or more of the output ports **426** through **432**; to monitor other I/O ports **414** for activity; to receive, decode, and act upon user commands received from controlling device **100** or input via front panel **420**; to display program guide information,

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listings of stored content, or setup and configuration menus at user request and act upon any resulting user input; etc.

Controlling devices **100** through **102** may incorporate device identity data into their command transmissions to STB **104**, for example without limitation as described in U.S. Pat. Nos. 7,266,701 or 7,167,913, both of common ownership. In summary, U.S. Pat. No. 7,266,701 describes a command transmission protocol wherein each data frame transmitted includes a 2-bit ID field, i.e. allowing for up to four separate controlling device identities; while U.S. Pat. No. 7,167,913 describes a command transmission protocol wherein each data frame transmitted includes a 5-bit TAG field, all or part of which may be utilized for controlling device identification, i.e. allowing for up to as many as 32 separate controlling device identities. For a more detailed description of the exact encoding schemes and frame formats used by these protocols, the interested reader may refer to the disclosures of those patents, both of which are incorporated herein by reference in their entirety.

As is known in the art, programming instructions stored in the memory of an STB (hereafter, "STB programming") may be responsive to, inter alia, receipt of a "guide" command to invoke an interactive program guide or a "menu" command to enable STB configuration and adjustment. In accordance with the teachings of this invention, the STB programming stored in memory **402** of an exemplary STB **104** for use in the system of FIG. 1 may be further enabled to adapt program guide and/or configuration GUIs and capabilities based upon the identity of the controlling device which issued the command to initiate that operation, as will now be described in conjunction with FIGS. 5a and 5b.

Upon receipt of a command from a controlling device, for example via remote control receiver **424**, a step **502** the STB programming may determine if the received command transmission constitutes a "guide" command. If so, at step **504** the STB programming may examine the controlling device identity information contained within that command transmission to determine if the transmission originated from children's remote **102**. If so at step **516** the operating mode of STB **104** may be set to "child" and at step **518** a program guide display suitable for interaction with a child may be displayed. By way of example, as illustrated in FIG. 8 such a program guide may comprise a simplified GUI **800** featuring colorful channel logos each corresponding to a single one of a group of direct-access buttons **802** provisioned on controlling device **102**. With reference to the operational mode setting of step **516**, by way of further explanation, in the exemplary implantation the setting of an operating mode in response to receipt of a controlling device identity embedded in a command such as "guide" or "menu" may be required to enable interpretation of subsequently received commands, for example GUI navigation inputs, to be consistent with the currently displayed GUI and/or the capabilities and features of the controlling device presently in use. As will become evident, this operating mode may be reset upon exit from the currently displayed GUI. As will also be appreciated, in other embodiments such functionality may be achieved through different means, for example a controlling device equipped with an accelerometer, capacitive touch sensor, or other such hardware may initiate a specific identification transmission upon initially detecting that it has been picked up by a user.

If the received device identity is not that of a children's remote, at step **506** the STB programming may next determine if the controlling device identity information indicates that the command transmission originated from a senior remote **101**. If so at step **512** the operating mode of STB **104**

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may be set to "senior" and at step **510** a program guide display adapted for use by senior citizens may be presented. By way of example, as illustrated in FIG. 7, such a guide display **700** may comprise a magnified or zoomed version of the standard guide **600** of FIG. 6, for greater legibility. Alternatively or in conjunction with zooming, in some embodiments the number of programming choices presented may also be reduced, for example limited to a preconfigured list of favorites.

If the received device identity is not that of a senior remote, the STB programming may then assume that the command transmission originated from a standard remote **100** (or, alternatively in some embodiments from an after-market universal remote control which does not support controlling device identity), set the operating mode of STB **104** to "standard" and present a default standard guide display, for example **600** as illustrated in FIG. 6.

Returning to step **502**, if the STB programming determines that the received command transmission does not constitute a "guide" command, then at step **520** it is next determined if the received command transmission constitutes a "menu" command. If so, at step **528** (FIG. 5b) the STB programming may examine the controlling device identity information contained within that command transmission to determine if the transmission originated from children's remote **102**. If so, the command is ignored and processing is completed at step **540**. By way of explanation, in the illustrative embodiment children's remote **102** may not be permitted to initiate STB configuration adjustments. It will be appreciated that while in some embodiments this may be achieved by simply not provisioning child's controlling device **102** with a "Menu" key, in other cases this may not be possible, for example where a single physical controlling device design may be adapted for various user populations by means of setup programming, jumpers, identity plug-in modules, etc.

If the received device identity is not that of a children's remote, at step **530** the STB programming may next determine if the controlling device identity information indicates that the command transmission originated from a senior remote **101**. If so, at step **532** the operating mode of STB **104** may be set to "senior" and a simplified STB configuration menu presented at step **534**. Such a menu may comprise, for example, only basic settings necessary for day-to-day use of STB **104**, and exclude adjustments which are chiefly required during initial setup or reconfiguration of STB **104**.

If the received device identity is not that of a children's remote or a senior remote, then at step **530** the STB programming may set the STB operating mode to "standard" and at step **538** may display a complete menu of STB configuration options. By way of example, options presented may include configuration of the audio/video output of STB **104** to match TV **106**, i.e. video resolution, picture aspect ratio, sound channels, etc.; setting of OSD timeouts and configurations for STB originated messages; selection of channel icons to be displayed while the STB is operating in "child" mode; etc. Returning to step **520** (FIG. 5a), if the received command is neither a "guide" nor a "menu" command, at step **522** the STB programming may next determine if the received command comprises a "select" or an "exit" command, i.e., a command which will result in the termination of an interactive menu or program guide GUI session. If so, at step **524** any currently set STB operating mode may be cleared prior to executing the received command at step **526**.

In a further exemplary embodiment as illustrated in FIG. 9, an appliance such as an AV receiver **902** which is

connected to and capable of switching between multiple source and sink devices such as TV set 106, cable STB 104, or DVD player 904 may be adapted to detect which one of several controlling devices 906 through 908, each corresponding to one of devices 104, 106 or 904, is currently in use and may configure itself accordingly—for example selecting appropriate inputs and outputs, setting audio volume and equalization levels, etc. While such an arrangement may be particularly advantageous when all devices involved are of the same manufacture and share a standardized command protocol and transmission format in order to facilitate controlling device identification, it will be appreciated that other techniques are also possible, for example AV receiver 902 may be provided with a universal IR receiver capable of identifying and decoding the command transmission formats of a multiplicity of manufactures, such as described for example in U.S. Pat. No. 7,042,366, incorporated herein by reference in its entirety.

While various concepts have been described in detail, it will be appreciated by those skilled in the art that various modifications and alternatives to those concepts could be developed in light of the overall teachings of the disclosure. For example, in an alternate embodiment, a consumer electronic device or system which is capable of operating in more than one mode, such as a gaming console which may also function as a DVD player; an Internet enabled TV which supports not only regular television viewing but also Web browsing, email, or IP telephony; etc.; may be provisioned with multiple controlling devices each optimized for a particular function, and may adapt video or audio parameters, user menus, input and output ports or formats, etc. in accordance with the controlling device currently in the possession of a user, i.e. upon being used and/or upon being picked up.

Further, while described in the context of functional modules and illustrated using block diagram format, it is to be understood that, unless otherwise stated to the contrary, one or more of the described functions and/or features may be integrated in a single physical device and/or a software module, or one or more functions and/or features may be implemented in separate physical devices or software modules. It will also be appreciated that a detailed discussion of the actual implementation of each module is not necessary for an enabling understanding of the invention. Rather, the actual implementation of such modules would be well within the routine skill of an engineer, given the disclosure herein of the attributes, functionality, and inter-relationship of the various functional modules in the system. Therefore, a person skilled in the art, applying ordinary skill, will be able to practice the invention set forth in the claims without undue experimentation. It will be additionally appreciated that the particular concepts disclosed are meant to be illustrative only and not limiting as to the scope of the invention which is to be given the full breadth of the appended claims and any equivalents thereof.

All patents cited within this document are hereby incorporated by reference in their entirety.

What is claimed is:

1. A computer implemented method performed by a multimedia access device having access to a plurality of media content, comprising:

in response to receiving a command communication for causing the multimedia access device to execute a commandable action from a mobile device:

determining if the command communication includes a device identifier data in addition to a command data and a system code data;

when it is determined that the command communication includes the device identifier:

using the device identifier in the command communication to identify a type for the mobile device;

identifying one or more preferences associated with the type for the mobile device;

receiving the one or more identified preferences;

configuring the multimedia access device based on the one or more identified preferences associated with the type for the mobile device; and

executing the commandable action; and

when it is determined that the command communication fails to include any device identifier, configuring the multimedia access device with a default set of preferences and executing the commandable action.

2. The method of claim 1, wherein configuring the multimedia access device further comprises adjusting visual display settings associated with a use of the multimedia access device.

3. The method of claim 1, wherein configuring the multimedia access device further comprises adjusting audio settings associated with a use of the multimedia access device.

4. The method of claim 1, wherein the mobile device comprises a remote control device.

5. The method of claim 1, wherein the mobile device uses an infrared communications protocol to communicate with the multimedia access device.

6. A non-transitory, computer-readable media having stored thereon instructions, the instructions, when executed by a multimedia access device having access to a plurality of media content, cause the multimedia access device to perform steps comprising:

in response to receiving a command communication for causing the multimedia access device to execute a commandable action from a mobile device:

determining if the command communication includes a device identifier data in addition to a command data and a system code data;

when it is determined that the command communication includes the device identifier:

using the device identifier in the command communication to identify a type for the mobile device;

identifying one or more preferences associated with the type for the mobile device;

receiving the one or more identified preferences;

configuring the multimedia access device based on the one or more identified preferences associated with the type for the mobile device; and

executing the commandable action; and

when it is determined that the command communication fails to include any device identifier, configuring the multimedia access device with a default set of preferences and executing the commandable action.

7. The non-transitory, computer-readable media of claim 6, wherein configuring the multimedia access device further comprises adjusting visual display settings associated with a use of the multimedia access device.

8. The non-transitory, computer-readable media of claim 6, wherein configuring the multimedia access device further comprises adjusting audio settings associated with a use of the multimedia access device.

9. The non-transitory, computer-readable media of claim 6, wherein the mobile device comprises a remote control device.

10. The non-transitory, computer-readable media of claim 6, wherein the mobile device uses an infrared communications protocol to communicate with the multimedia access device.

11. The non-transitory, computer-readable media of claim 6, wherein the multimedia access device comprises a set-top box. 5

12. The non-transitory, computer-readable media of claim 6, wherein the multimedia access device comprises a television. 10

13. The non-transitory, computer-readable media of claim 6, wherein the selected set of media access options comprises an electronic program guide.

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